Math 3A Problem Set 5 Derivative Applications Name Wes Oakley

Please make sure that you show all of the steps, not just the answer.

1) A 13-foot ladder is leaning against a house when its base begins to slide away. By the time the base is 12 feet from the building, the base is moving at 5 ft/sec.

a) How fast is the top of the ladder sliding down the wall then?

x is the distance to the base of the wall.
y is the height of the top of the ladder on the wall.
Ladder length is constant at 13.

$$x^2 + y^2 = 13^2$$

$$x^2 + y^2 = 169$$

Differentiate with respect to time

$$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(169) \rightarrow \boxed{2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0}$$

Given $x = 12$, $\frac{dx}{dt} = 5$

$$(12^2) + y^2 = 169$$

$$144 + y^2 = 169$$

$$y^2 = 25$$

$$y = 5$$

Sub into derivative

$$2(12)(5) + 2(5)\frac{dy}{dt} = 0$$

$$120 + 10\frac{dy}{dt} = 0$$

$$10\frac{dy}{dt} = -120$$

$$\frac{dy}{dt} = -12 \rightarrow \boxed{12 \frac{ft}{sec}}$$

b) At what rate is the area of the triangle formed by the ladder changing then?

The triangle area is $A = \frac{1}{2}xy$

↳ we want $\frac{dA}{dt}$

$$\frac{dA}{dt} = \frac{1}{2}\frac{d}{dt}(xy)$$

$$\frac{dA}{dt} = \frac{1}{2}\left(x\frac{dy}{dt} + y\frac{dx}{dt}\right)$$

$x = 12$, $y = 5$, $\frac{dx}{dt} = 5$, $\frac{dy}{dt} = -12$

$$\frac{dA}{dt} = \frac{1}{2}(12(-12) + 5(5))$$

$$= \frac{1}{2}(-144 + 25)$$

$$= \frac{1}{2}(-119)$$

$$\frac{dA}{dt} = -\frac{119}{2} \text{ or } \boxed{-59.5 \ ft^2/sec}$$

c) At what rate is the angle θ between the ladder and the ground changing then?

$\cos\theta = \frac{x}{13}$ because for that angle, adjacent $= x$ and hypotenuse $= 13$

$$\frac{d}{dt}(\cos\theta) = \frac{d}{dt}\left(\frac{x}{13}\right)$$

$$-\sin\theta\frac{d\theta}{dt} = \frac{1}{13}\frac{dx}{dt}$$

$$\frac{d\theta}{dt} = -\frac{\frac{1}{13}\frac{dx}{dt}}{\sin\theta}$$

we know $dx/dt = 5$
and $\sin\theta = \frac{y}{h} \rightarrow y = 5, h = 13 \rightarrow \frac{5}{13}$

$$\frac{d\theta}{dt} = -\frac{\frac{1}{13}(5)}{\left(\frac{5}{13}\right)} \Rightarrow -\frac{5/13}{5/13} \Rightarrow \boxed{-1 \ Rad/sec}$$

2) On December 20 of last year, the Mega Millions Jackpot was \$465 million and by December 27 it had grown to \$565 million. Let $A(t) = Pe^{rt}$ be the function that represents the value of the Mega Millions Jackpot where $t = 0$ represents December 20.

a) Find the formula for $A(t)$ and then find $A(10)$ and explain what it means in this situation.

$A(t) = Pe^{rt}$

Jackpot at $t=0$ so,

$A(0) = 465$

$465 = Pe^{r(0)}$

$465 = Pe^{0}$

$465 = P(1)$

$P = 465$

$A(t) = 465e^{rt}$

December 27th is 7 days after the 20th, so $t = 7$

↳ and Jackpot is 565 million.

$A(7) = 565$

$565 = 465e^{7r}$

$\frac{565}{465} = e^{7r}$

$\frac{113}{93} = e^{7r}$

$\ln\left(\frac{113}{93}\right) = \ln\left(e^{7r}\right)$

$\ln\left(\frac{113}{93}\right) = 7r$

$r = \frac{1}{7}\ln\left(\frac{113}{93}\right)$

$A(t) = 465e^{(\frac{1}{7}\ln(113/93)t}$

$A(10) = 465e^{(\frac{10}{7}\ln(113/93))}$

$\boxed{A(10) \approx 614.7 \text{ million}}$

this means that 10 days after dec. 20 the Jackpot is predicted to be about \$614.7 million

b) Find $A'(10)$ and explain what it means in this situation.

$A(t) = 465e^{rt}$

$A'(t) = 465re^{rt}$

$A'(10) = 465re^{r(10)}$

$r = \frac{1}{7}\ln\left(\frac{113}{93}\right) \approx 0.0279$

$A'(10) = 465\left(\frac{1}{7}\ln\left(\frac{113}{93}\right)\right)e^{10(1/7\ln(113/93))}$

$465e^{10r} = A(10)$ So...

$A'(10) = rA(10)$

$A'(10) \approx (0.027916)(614.7)$

$\boxed{A'(10) \approx 17.16 \text{ million / day}}$

this means that 10 days after Dec. 20 the Jackpot is increasing at about \$17.16 million per day.

3) Water is being stored in a right circular cylinder that has a radius of 18 inches. How fast is the water level falling if water is being drained out at a rate of 300 cubic inches per second.

$V =$ volume of water

$h =$ height of water

Cylinder has a constant radius: $r = 18$

Volume of the cylinder: $V = \pi r^2 h$

$V = \pi(18)^2 h$

$V = 324\pi h$

water is being drained, so the volume is decreasing $\frac{dV}{dt} = -300$

$\frac{dV}{dt} = 324\pi \frac{dh}{dt}$

$(-300) = 324\pi \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{-300}{324\pi} = \frac{-25}{27\pi}$

$\boxed{\frac{dh}{dt} \approx -0.295 \text{ in/sec}}$